

TrES-5b

R-0831

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_181207 · created 2026-07-28T17:14:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458459.6204 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.015 +/- 0.053
Transit depth (Rp/Rs)^2	0.022 +/- 0.16 [%]
Orbital Inclination (inc)	85.45 +/- 2.91
Airmass coefficient 1 (a1)	1.06 +/- 0.94
Airmass coefficient 2 (a2)	0.03 +/- 0.75
Scatter in the residuals of the lightcurve fit is	100.95 %
Best Comparison Star	#3 - [49, 138]
Optimal Aperture	2.87
Optimal Annulus	10.66
Transit Duration (day)	0.064 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.015 ± 0.053 vs 0.142 (deviation 2.4σ)
Duration fit vs published	0.064 d vs 0.0758 d (deviation 0.62σ)
Mid-time O–C	-6.8 min
Depth SNR	0.3
Residual scatter	100.95 %

Light curve

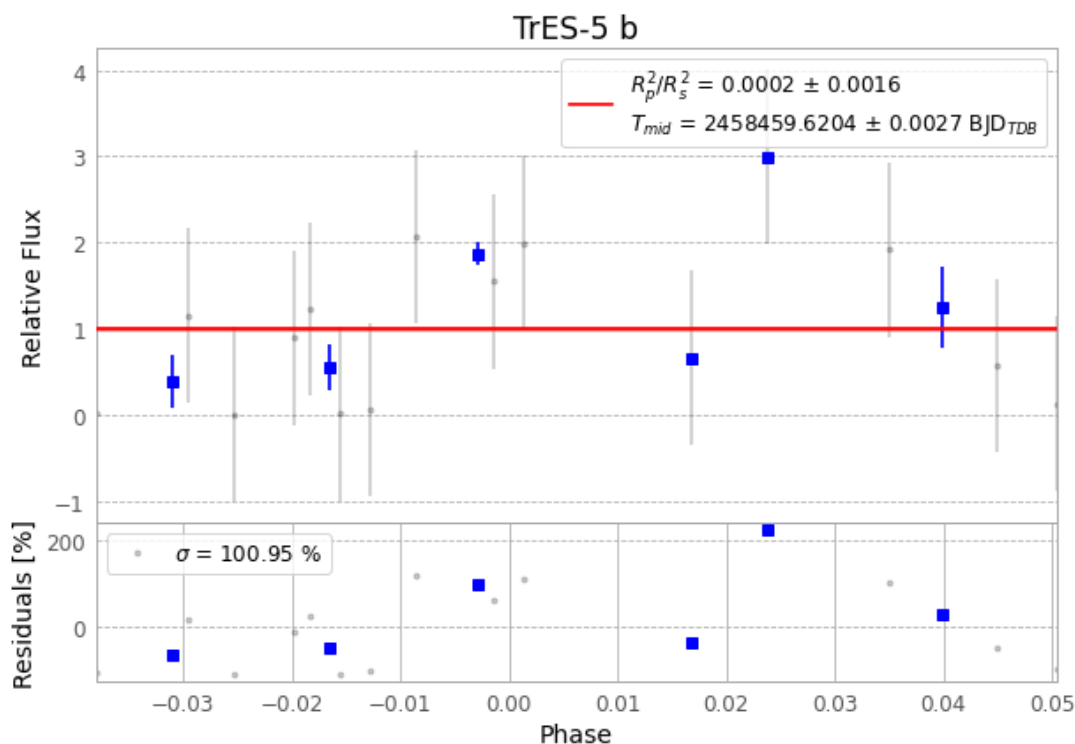


Figure 1 — FinalLightCurve_TrES-5 b_2018-12-06.png (SHA-256 255b328c48884045...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 405], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 34eece12f65d9d8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

66 FITS frames (43 MB), TRES-5181207013010.FITS → TRES-5181207044812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-181207.log (9d466972a35c...), FinalLightCurve_TrES-5 b_2018-12-06.png (255b328c4888...), FinalLightCurve_TrES-5 b_2018-12-06.csv (2c129bdd448c...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 17:14Z.